STATISTICAL TABLES

Cumulative normal distribution

Critical values of the t distribution

Critical values of the F distribution

Critical values of the chi-squared distribution

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Table A.1

Cumulative Standardized Normal Distribution

A(z) is the integral of the standardized normal distribution from -inf. to z (in other words, the area under the curve to the left of z). It gives the probability of a normal random variable not being more than z standard deviations above its mean. Values of z of particular importance:

1.645	0.9500	Lower limit of right 5% tail
1.960	0.9750	Lower limit of right 2.5% tail
2.326	0.9900	Lower limit of right 1% tail
2.576	0.9950	Lower limit of right 0.5% tail
3.090	0.9990	Lower limit of right 0.1% tail
3.291	0.9995	Lower limit of right 0.05% tail

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998
3.5	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998
3.6	0.9998									

Table A.2

t Distribution: Critical Values of t

		Significance level					
Degrees of freedom	*Two-tailed test:*	10%	5%	2%	1%	0.2%	0.1%
	One-tailed test:	5%	2.5%	1%	0.5%	0.1%	0.05%
1		6.314	12.706	31.821	63.657	318.309	636.619
2		2.920	4.303	6.965	9.925	22.327	31.599
3		2.353	3.182	4.541	5.841	10.215	12.924
4		2.132	2.776	3.747	4.604	7.173	8.610
5		2.015	2.571	3.365	4.032	5.893	6.869
6		1.943	2.447	3.143	3.707	5.208	5.959
7		1.894	2.365	2.998	3.499	4.785	5.408
8		1.860	2.306	2.896	3.355	4.501	5.041
9		1.833	2.262	2.821	3.250	4.297	4.781
10		1.812	2.228	2.764	3.169	4.144	4.587
11		1.796	2.201	2.718	3.106	4.025	4.437
12		1.782	2.179	2.681	3.055	3.930	4.318
13		1.771	2.160	2.650	3.012	3.852	4.221
14		1.761	2.145	2.624	2.977	3.787	4.140
15		1.753	2.131	2.602	2.947	3.733	4.073
16		1.746	2.120	2.583	2.921	3.686	4.015
17		1.740	2.110	2.567	2.898	3.646	3.965
18		1.734	2.101	2.552	2.878	3.610	3.922
19		1.729	2.093	2.539	2.861	3.579	3.883
20		1.725	2.086	2.528	2.845	3.552	3.850
21		1.721	2.080	2.518	2.831	3.527	3.819
22		1.717	2.074	2.508	2.819	3.505	3.792
23		1.714	2.069	2.500	2.807	3.485	3.768
24		1.711	2.064	2.492	2.797	3.467	3.745
25		1.708	2.060	2.485	2.787	3.450	3.725
26		1.706	2.056	2.479	2.779	3.435	3.707
27		1.703	2.052	2.473	2.771	3.421	3.690
28		1.701	2.048	2.467	2.763	3.408	3.674
29		1.699	2.045	2.462	2.756	3.396	3.659
30		1.697	2.042	2.457	2.750	3.385	3.646
40		1.684	2.021	2.423	2.704	3.307	3.551
50		1.676	2.009	2.403	2.678	3.261	3.496
60		1.671	2.000	2.390	2.660	3.232	3.460
80		1.664	1.990	2.374	2.639	3.195	3.416
100		1.660	1.984	2.364	2.626	3.174	3.390
120		1.658	1.980	2.358	2.617	3.160	3.373
200		1.653	1.972	2.345	2.601	3.131	3.340
500		1.648	1.965	2.334	2.586	3.107	3.310
inf.		1.645	1.960	2.326	2.576	3.090	3.291

Table A.3

F Distribution: Critical Values of F (5% significance level)

v2 \ v1	1	2	3	4	5	6	8	10	12	16	20
1	161.45	199.50	215.71	224.58	230.16	233.99	238.88	241.88	243.91	246.46	248.01
2	18.51	19.00	19.16	19.25	19.30	19.33	19.37	19.40	19.41	19.43	19.45
3	10.13	9.55	9.28	9.12	9.01	8.94	8.85	8.79	8.74	8.69	8.66
4	7.71	6.94	6.59	6.39	6.26	6.16	6.04	5.96	5.91	5.84	5.80
5	6.61	5.79	5.41	5.19	5.05	4.95	4.82	4.74	4.68	4.60	4.56
6	5.99	5.14	4.76	4.53	4.39	4.28	4.15	4.06	4.00	3.92	3.87
8	5.32	4.46	4.07	3.84	3.69	3.58	3.44	3.35	3.28	3.20	3.15
10	4.96	4.10	3.71	3.48	3.33	3.22	3.07	2.98	2.91	2.83	2.77
12	4.75	3.89	3.49	3.26	3.11	3.00	2.85	2.75	2.69	2.60	2.54
15	4.54	3.68	3.29	3.06	2.90	2.79	2.64	2.54	2.48	2.38	2.33
20	4.35	3.49	3.10	2.87	2.71	2.60	2.45	2.35	2.28	2.18	2.12
30	4.17	3.32	2.92	2.69	2.53	2.42	2.27	2.16	2.09	1.99	1.93
40	4.08	3.23	2.84	2.61	2.45	2.34	2.18	2.08	2.00	1.90	1.84
60	4.00	3.15	2.76	2.53	2.37	2.25	2.10	1.99	1.92	1.82	1.75
120	3.92	3.07	2.68	2.45	2.29	2.18	2.02	1.91	1.83	1.73	1.66
inf.	3.84	3.00	2.60	2.37	2.21	2.10	1.94	1.83	1.75	1.64	1.57

Table A.3 (continued)

F Distribution: Critical Values of F (5% significance level)

v2 \ v1	25	30	40	50	60	75	100	150	200
1	249.26	250.10	251.14	251.77	252.20	252.62	253.04	253.46	253.68
2	19.46	19.46	19.47	19.48	19.48	19.48	19.49	19.49	19.49
3	8.63	8.62	8.59	8.58	8.57	8.56	8.55	8.54	8.54
4	5.77	5.75	5.72	5.70	5.69	5.68	5.66	5.65	5.65
5	4.52	4.50	4.46	4.44	4.43	4.42	4.41	4.39	4.39
10	2.73	2.70	2.66	2.64	2.62	2.60	2.59	2.57	2.56
15	2.28	2.25	2.20	2.18	2.16	2.14	2.12	2.10	2.10
20	2.07	2.04	1.99	1.97	1.95	1.93	1.91	1.89	1.88
30	1.88	1.84	1.79	1.76	1.74	1.72	1.70	1.67	1.66
60	1.69	1.65	1.59	1.56	1.53	1.51	1.48	1.45	1.44
100	1.62	1.57	1.52	1.48	1.45	1.42	1.39	1.36	1.34
inf.	1.52	1.47	1.41	1.36	1.33	1.30	1.26	1.22	1.19

Table A.3

F Distribution: Critical Values of F (1% significance level)

v2 \ v1	1	2	3	4	5	6	8	10	12	16	20
1	161.45	199.50	215.71	224.58	230.16	233.99	238.88	241.88	243.91	246.46	248.01
2	18.51	19.00	19.16	19.25	19.30	19.33	19.37	19.40	19.41	19.43	19.45
3	10.13	9.55	9.28	9.12	9.01	8.94	8.85	8.79	8.74	8.69	8.66
4	7.71	6.94	6.59	6.39	6.26	6.16	6.04	5.96	5.91	5.84	5.80
5	6.61	5.79	5.41	5.19	5.05	4.95	4.82	4.74	4.68	4.60	4.56
6	5.99	5.14	4.76	4.53	4.39	4.28	4.15	4.06	4.00	3.92	3.87
8	5.32	4.46	4.07	3.84	3.69	3.58	3.44	3.35	3.28	3.20	3.15
10	4.96	4.10	3.71	3.48	3.33	3.22	3.07	2.98	2.91	2.83	2.77
12	4.75	3.89	3.49	3.26	3.11	3.00	2.85	2.75	2.69	2.60	2.54
15	4.54	3.68	3.29	3.06	2.90	2.79	2.64	2.54	2.48	2.38	2.33
20	4.35	3.49	3.10	2.87	2.71	2.60	2.45	2.35	2.28	2.18	2.12
30	4.17	3.32	2.92	2.69	2.53	2.42	2.27	2.16	2.09	1.99	1.93
40	4.08	3.23	2.84	2.61	2.45	2.34	2.18	2.08	2.00	1.90	1.84
60	4.00	3.15	2.76	2.53	2.37	2.25	2.10	1.99	1.92	1.82	1.75
120	3.92	3.07	2.68	2.45	2.29	2.18	2.02	1.91	1.83	1.73	1.66
inf.	3.84	3.00	2.60	2.37	2.21	2.10	1.94	1.83	1.75	1.64	1.57

Table A.3 (continued)

F Distribution: Critical Values of F (1% significance level)

v2 \ v1	25	30	40	50	60	75	100	150	200
1	249.26	250.10	251.14	251.77	252.20	252.62	253.04	253.46	253.68
2	19.46	19.46	19.47	19.48	19.48	19.48	19.49	19.49	19.49
3	8.63	8.62	8.59	8.58	8.57	8.56	8.55	8.54	8.54
4	5.77	5.75	5.72	5.70	5.69	5.68	5.66	5.65	5.65
5	4.52	4.50	4.46	4.44	4.43	4.42	4.41	4.39	4.39
10	2.73	2.70	2.66	2.64	2.62	2.60	2.59	2.57	2.56
15	2.28	2.25	2.20	2.18	2.16	2.14	2.12	2.10	2.10
20	2.07	2.04	1.99	1.97	1.95	1.93	1.91	1.89	1.88
30	1.88	1.84	1.79	1.76	1.74	1.72	1.70	1.67	1.66
60	1.69	1.65	1.59	1.56	1.53	1.51	1.48	1.45	1.44
100	1.62	1.57	1.52	1.48	1.45	1.42	1.39	1.36	1.34
inf.	1.52	1.47	1.41	1.36	1.33	1.30	1.26	1.22	1.19

Table A.3

F Distribution: Critical Values of F (0.1% significance level)

v2 \ v1	1	2	3	4	5	6	8	10	12	16	20
1	161.45	199.50	215.71	224.58	230.16	233.99	238.88	241.88	243.91	246.46	248.01
2	18.51	19.00	19.16	19.25	19.30	19.33	19.37	19.40	19.41	19.43	19.45
3	10.13	9.55	9.28	9.12	9.01	8.94	8.85	8.79	8.74	8.69	8.66
4	7.71	6.94	6.59	6.39	6.26	6.16	6.04	5.96	5.91	5.84	5.80
5	6.61	5.79	5.41	5.19	5.05	4.95	4.82	4.74	4.68	4.60	4.56
6	5.99	5.14	4.76	4.53	4.39	4.28	4.15	4.06	4.00	3.92	3.87
8	5.32	4.46	4.07	3.84	3.69	3.58	3.44	3.35	3.28	3.20	3.15
10	4.96	4.10	3.71	3.48	3.33	3.22	3.07	2.98	2.91	2.83	2.77
12	4.75	3.89	3.49	3.26	3.11	3.00	2.85	2.75	2.69	2.60	2.54
15	4.54	3.68	3.29	3.06	2.90	2.79	2.64	2.54	2.48	2.38	2.33
20	4.35	3.49	3.10	2.87	2.71	2.60	2.45	2.35	2.28	2.18	2.12
30	4.17	3.32	2.92	2.69	2.53	2.42	2.27	2.16	2.09	1.99	1.93
40	4.08	3.23	2.84	2.61	2.45	2.34	2.18	2.08	2.00	1.90	1.84
60	4.00	3.15	2.76	2.53	2.37	2.25	2.10	1.99	1.92	1.82	1.75
120	3.92	3.07	2.68	2.45	2.29	2.18	2.02	1.91	1.83	1.73	1.66
inf.	3.84	3.00	2.60	2.37	2.21	2.10	1.94	1.83	1.75	1.64	1.57

Table A.3 (continued)

F Distribution: Critical Values of F (0.1% significance level)

v2 \ v1	25	30	40	50	60	75	100	150	200
1	249.26	250.10	251.14	251.77	252.20	252.62	253.04	253.46	253.68
2	19.46	19.46	19.47	19.48	19.48	19.48	19.49	19.49	19.49
3	8.63	8.62	8.59	8.58	8.57	8.56	8.55	8.54	8.54
4	5.77	5.75	5.72	5.70	5.69	5.68	5.66	5.65	5.65
5	4.52	4.50	4.46	4.44	4.43	4.42	4.41	4.39	4.39
10	2.73	2.70	2.66	2.64	2.62	2.60	2.59	2.57	2.56
15	2.28	2.25	2.20	2.18	2.16	2.14	2.12	2.10	2.10
20	2.07	2.04	1.99	1.97	1.95	1.93	1.91	1.89	1.88
30	1.88	1.84	1.79	1.76	1.74	1.72	1.70	1.67	1.66
60	1.69	1.65	1.59	1.56	1.53	1.51	1.48	1.45	1.44
100	1.62	1.57	1.52	1.48	1.45	1.42	1.39	1.36	1.34
inf.	1.52	1.47	1.41	1.36	1.33	1.30	1.26	1.22	1.19

Table A.4

Chi-Squared Distribution: Critical Values of Chi-Squared

	Significance level				
Degrees of freedom	**10%**	**5%**	**2.5%**	**1%**	**0.1%**
1	2.706	3.841	5.024	6.635	10.828
2	4.605	5.991	7.378	9.210	13.816
3	6.251	7.815	9.348	11.345	16.266
4	7.779	9.488	11.143	13.277	18.467
5	9.236	11.070	12.833	15.086	20.515
6	10.645	12.592	14.449	16.812	22.458
7	12.017	14.067	16.013	18.475	24.322
8	13.362	15.507	17.535	20.090	26.124
9	14.684	16.919	19.023	21.666	27.877
10	15.987	18.307	20.483	23.209	29.588
12	18.549	21.026	23.337	26.217	32.909
15	22.307	24.996	27.488	30.578	37.697
20	28.412	31.410	34.170	37.566	45.315
25	34.382	37.652	40.646	44.314	52.620
30	40.256	43.773	46.979	50.892	59.703
40	51.805	55.758	59.342	63.691	73.402
50	63.167	67.505	71.420	76.154	86.661
60	74.397	79.082	83.298	88.379	99.607
80	96.578	101.879	106.629	112.329	124.839
100	118.498	124.342	129.561	135.807	149.449